# A Hybrid Fuzzy Logic–Based Intelligent Traffic Signal Control with Optimization Mechanism

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***Abstract-Fixed-time traffic lights are still used in many intersections, although this method is not always suitable when the traffic condition changes from one period to another. When the signal timing is too rigid, vehicles may wait even when the road is not crowded, while queues can become longer during busy conditions. This work discusses a Smart Traffic Light Control System using Fuzzy Logic. The system uses two input variables, namely the waiting time and number of vehicles, to decide the green-light duration. These inputs are converted into linguistic categories and processed using fuzzy rules that follow common traffic-control reasoning. The evaluation shows that the system gives different green-light decisions for different traffic situations. The waiting-time values used in the evaluation include 15 seconds for a short condition, 60 seconds for a medium condition, and 90 seconds for a long condition. From these results, the proposed system shows that Fuzzy Logic can support a more flexible signal-control process than a fixed-time approach. This study contributes by combining vehicle volume and waiting time in one fuzzy decision framework for intersection management. Future work can continue this study through real-time testing with IoT-based traffic sensors and integration with intelligent transportation systems.***

***Keywords—artificial intelligence, fuzzy logic, smart traffic light, intelligent transportation system***

## I. INTRODUCTION

Urban traffic is becoming more difficult to manage because the number of vehicles keeps increasing, while road capacity does not grow at the same pace. Traffic lights are one of the main tools used to control movement at intersections. However, many traffic light systems still use fixed-time schedules, so they cannot adjust directly to the condition on the road [1]. This can create problems in both quiet and crowded situations. When the traffic volume is low, drivers may still have to wait for a long red light. On the other hand, when the traffic volume is high, the available green time may not be enough to reduce the queue. Because of this, adaptive traffic signal control is needed so that the signal can respond more closely to actual traffic demand. Better signal control can help improve intersection performance, reduce vehicle delay, and support smoother traffic flow [2]. Artificial Intelligence is often used for problems that need flexible decision-making. One AI method that fits this case is Fuzzy Logic, because traffic data is not always exact or easy to separate into fixed categories. For example, traffic density can be described as low, medium, or high without using a single sharp boundary [3]. This makes Fuzzy Logic useful for traffic-control decisions because it can represent uncertain conditions in a way that is more like to human reasoning. Conventional signal-control methods based on fixed timing are less able to handle sudden changes in vehicle volume, different transport modes, driver behavior, and pedestrian movement. Several recent studies have also explored Large Language Models (LLMs) as traffic-control agents. These models may support interpretation and decision-making in dynamic traffic situations [4]. With real-time feedback, a traffic controller can improve throughput and respond more effectively to changing intersection conditions [5]. Even so, many existing systems still depend on fixed timing or limited sensing. As a result, they may not capture traffic patterns accurately on busy roads, and pedestrian needs are not always considered properly.

Previous research has tried to solve these problems using different approaches. Some studies use image processing to observe the traffic condition and support the traffic-light decision process. By using camera data, the system can estimate the road condition more directly and may also improve pedestrian accessibility [6][7]. In image-processing-based systems, Canny Edge Detection is one method that has been used to detect and analyze traffic flow [8][9]. Other studies discuss recommendation systems for urban traffic control, especially because they may improve control efficiency and support larger-scale implementation [10]. However, these systems can become complicated because they involve many connected components. In addition, studies that discuss this topic in a systematic and complete way are still limited.

Based on this background, this study focuses on the design and analysis of a traffic light control system using Fuzzy Logic. The system considers vehicle density and waiting time to determine the green-signal duration. Instead of using the same timing for every condition, the proposed approach changes the signal duration based on the input condition. This makes the system more adaptive than a conventional fixed-time traffic light. The utilization of Fuzzy Logic is expected to help improve traffic flow and reduce congestion in urban intersections. The rest of this paper is organized as follows. Section II helps to understand the Fuzzy Logic-based traffic

light control system. Section III comprehends results and discussion. Section IV presents the conclusion.

## II. Fuzzy Logic-Based Traffic Light Control System

Fuzzy Logic is a computational method that does not only use true or false values. In classical binary logic, a condition is usually represented by 1 or 0. In Fuzzy Logic, a condition can have a degree of membership between 0 and 1 [11]. This is useful for problems that are uncertain or not easy to define with exact limits. Traffic conditions are a good example because an intersection is not always simply empty or crowded. Sometimes the condition is between two categories. By using Fuzzy Logic, different levels of traffic density and congestion can be included in the decision process. This makes the method more flexible for changing traffic conditions than a strict binary approach [12][13].

### *A. System Inputs and Output*

The proposed system utilizes one output variable and two input variables The input variables describe the traffic condition, while the output variable gives the signal decision. In this system, the inputs are combined to produce one green-light duration. The variables are listed as follows:

- Condition : Low, Medium, High
- Waiting Time: Short, Medium, Long
- Green Light Duration: Short, Normal, Long

These linguistic variables make the traffic condition easier to interpret. Instead of reading only numerical values, the system can describe the condition using terms that are easier to understand, such as low, medium, and high [14]. This helps convert traffic data into a form that can be used in the fuzzy decision process.

### *B. Fuzzy Rule Base*

The control process uses fuzzy rules based on simple traffic-control logic. Some of the main rules are written as follows:

- If the number of vehicles is high and the waiting time is long, then green light duration is long.
- If the number of vehicles is medium and the waiting time is medium, then green light duration is normal.
- If the number of vehicles is low and the waiting time is short, then green light duration is short.
- If the number of vehicles is high and the waiting time is short, then green light duration is normal.

These rules help the system decide the signal duration in a way that is similar to basic human reasoning. If the queue is large and vehicles have waited for a long time, the green duration should be longer. If the road is quiet and the waiting time is short, the green duration can be shorter [15][16].

### *C. Fuzzy Inference Process*

The Fuzzy Logic control system has three main stages. These stages show how the input data is changed into the final green-light duration.

1. Fuzzification: crisp input values are converted into fuzzy membership values.
2. Inference Engine: fuzzy rules are implemented to generate fuzzy outputs.
3. Defuzzification: fuzzy outputs are converted into a crisp green-light duration.

### *D. Definition of Variables*

1. Input Variable 1: Number of Vehicles (NV)

   Let:

   - $NV$ = number of vehicles detected at the intersection

   Linguistic terms:

   - Low
   - Medium
   - High

   Therefore, the universe of discourse is :

   $$NV \in [0,30]$$

2. Input Variable 2: Waiting Time (WT)

   Let:

   - $WT$= average waiting time (seconds)

   Linguistic terms:

   - Short
   - Medium
   - Long

   Universe of discourse is presented as

   $$WT \in [0,120]$$

3. Output Variable: Green Light Duration (GD)

   Let:

   - $GD$*= green light duration (seconds)*

   Linguistic terms:

   - Short
   - Normal
   - Long

   Universe of discourse is presented as

   $$GD \in [10,60]$$

### *E. Membership Function*

For simplicity and clarity, triangular membership functions are used as follows.

1. Number of Vehicles

$$\mu_{\text{Low}}(NV) = \begin{cases} 1, & NV \leq 5 \\ \dfrac{10 - NV}{5}, & 5 < NV < 10 \\ 0, & NV \geq 10 \end{cases}$$

$$\mu_{\text{Medium}}(NV) = \begin{cases} 0, & NV \le 5 \\ \dfrac{NV-5}{10}, & 5 < NV < 15 \\ \dfrac{25-NV}{10}, & 15 \le NV < 25 \\ 0, & NV \ge 25 \end{cases}$$

$$\mu_{\text{High}}(NV) = \begin{cases} 0, & NV \le 20 \\ \dfrac{NV-20}{10}, & 20 < NV < 30 \\ 1, & NV \ge 30 \end{cases}$$

**2.** Waiting Time

$$\mu_{\text{Short}}(WT) = \begin{cases} 1, & WT \le 20 \\ \dfrac{40-WT}{20}, & 20 < WT < 40 \\ 0, & WT \ge 40 \end{cases}$$

$$\mu_{\text{Medium}}(WT) = \begin{cases} 0, & WT \le 30 \\ \dfrac{WT-30}{30}, & 30 < WT < 60 \\ \dfrac{90-WT}{30}, & 60 \le WT < 90 \\ 0, & WT \ge 90 \end{cases}$$

$$\mu_{\text{Long}}(WT) = \begin{cases} 0, & WT \le 60 \\ \dfrac{WT-60}{30}, & 60 < WT < 90 \\ 1, & WT \ge 90 \end{cases}$$

## III. Result and Discussion

This section explains the analysis and results of the proposed Fuzzy Logic-based smart traffic light control system. The discussion focuses on how the system changes the green-light duration based on the traffic input. It also compares the idea of adaptive control with the fixed-time traffic signal approach.

In the fuzzification stage, crisp input values from the traffic sensors are changed into fuzzy membership values. This allows the system to represent traffic conditions using linguistic variables. Vehicle density is described as low, medium, or high, while waiting time is described as short, medium, or long. The result of this stage shows that one input value can belong to more than one fuzzy set at the same time. This usually happens when the value is close to the boundary between two categories. Because of this overlap, the system can handle uncertain conditions more smoothly.

For example, a traffic condition near the boundary between low and medium may be treated as partly low and partly medium. This is closer to real traffic conditions, where the situation is not always exact. The fuzzy inference process in this study uses the Mamdani method. In this method, the system checks the input variables against the rule base and determines which rules are active. Each active rule has a firing strength, which shows how much that rule affects the final decision.

The inference process can be observed from several traffic scenarios. When number of vehicles is high and waiting time is long, the rule for a longer green light becomes stronger. When number of vehicles is low and waiting time is short, the system gives a shorter green-light interval. For a mixed condition, such as high vehicle density but short waiting time, the system gives a normal green duration so that the traffic flow remains balanced.

By using vehicle density and waiting time together, the system can make a more balanced decision. The fuzzy outputs from the active rules are combined, and the final crisp value is obtained using the centroid defuzzification method. This method gives a clear green-light duration from the fuzzy output. The results show that the green-light decisions are reasonable for the tested traffic conditions. In a low-traffic situation, the system gives a shorter green duration, so unnecessary waiting can be reduced. In a heavy-traffic situation, the system gives a longer green duration to help reduce the queue. Compared with a fixed-time system, this approach is more flexible because the decision changes according to the input condition. Table I presents the evaluation results of the proposed Fuzzy Logic-based smart traffic light control system under different traffic conditions.

TABLE I. Fuzzy Input-Output Evaluation for Smart Traffic Light System

| No | Number of Vehicles (NV) | Waiting Time (WT) (s) | Activated Rule | Firing Strength ($\alpha$) | Green Duration Decision |
|---|---|---|---|---|---|
| 1. | Low (5) | Short (15) | Rule 3 | 0.80 | Short (≈ 20 s) |
| 2 | Medium (15) | Medium (60) | Rule 2 | 0.70 | Normal (≈ 35 s) |
| 3 | High (25) | Long (90) | Rule 1 | 0.90 | Long (≈ 55 s) |
| 4 | High (22) | Short (25) | Rule 4 | 0.20 | Normal (approximately 35 s) |

[a.] Firing strength (alpha) is calculated using the MIN operator in the Mamdani fuzzy inference method.

Table I shows how the proposed system corresponds to different traffic conditions. Each row represents one scenario utilizing number of vehicles of vehicles and waiting time at the intersection. The shortest waiting time in the table is 15 seconds, the medium waiting time is 60 seconds, and the longest waiting time is 90 seconds. Number of Vehicles (NV) represents traffic density detected by sensors, while Waiting Time (WT) represents the average waiting time in seconds. The Activated Rule column shows which rule becomes dominant in the decision process. Firing Strength (alpha) shows the level of rule activation, and Green Duration Decision shows the final green-light duration after defuzzification.

## IV. Conclusion

This paper presents a Fuzzy Logic-based smart traffic light control system as an alternative to conventional fixed-time traffic signals. The proposed system considers number of vehicles and waiting time before deciding the green-light duration. Based on the evaluation, the system can give different decisions for different traffic conditions. The tested waiting-time values include 15 seconds, 60 seconds, and 90 seconds, which represent short, medium, and long waiting conditions. The result indicates that Fuzzy Logic can support

a more adaptive signal-control process because it does not depend on one fixed timing pattern. Overall, the proposed system can help improve traffic efficiency and reduce congestion at intersections. Future development can focus on real-time implementation using IoT-based sensors and integration with intelligent transportation systems for smarter urban mobility.

## AUTHOR CONTRIBUTORSHIP

Keisa Putri Alisa led the development of the research concept and designed the methodology. Marcellino Varian Saputra conducted the formal analysis and prepared the initial draft of the manuscript. Shieren Revalina Jesslyn was responsible for collecting the data and contributed to refining the research methodology. Selvi Lukman supervised the overall project, provided critical feedback, and contributed to reviewing and editing the manuscript to improve its intellectual content.

## DATA AVAILABILITY

The utilized data in this study are publicly available at https://zenodo.org/records/18144407.

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